

When Does Streaming Tool Use Help?

Characterizing Tool-Intent Stabilization in Streaming Retrieval-Augmented Generation

Anonymous ACL submission

Abstract

Streaming Retrieval-Augmented Generation (Streaming RAG) reduces user-perceived latency by issuing tool queries in parallel with ongoing user input, before the utterance is complete. Reported gains are aggregate, yet the mechanism’s benefit is fundamentally *query-intrinsic*: speculation can only help when the correct tool query becomes determinable before the user stops speaking or typing. We isolate and measure this property—**tool-intent stabilization**, the point in the input stream at which a speculative query’s retrieval converges to the answer-bearing result. On the CRAG benchmark (1371 validation questions) we (i) measure the distribution of stabilization, (ii) derive a model-agnostic bound H on the portion of tool latency that can be hidden behind the user’s remaining input, as a function of tool latency L and input cadence δ , (iii) validate it against a working streaming pipeline—where realized savings meet or exceed the bound *in aggregate*, establishing H as a conservative aggregate floor rather than a per-query predictor (its per-query rank correlation with measured savings is near zero, §5.4)—and (iv) identify which query properties predict early versus late stabilization. The study requires no model training and runs on commodity CPU hardware. All experiments use CRAG (English), BM25 retrieval, and a uniform-word-stream proxy for input cadence; speech is the motivating application but is not directly tested. We report latency hiding decomposed rather than blended. At a realistic operating point ($L=600$ ms, $\delta=3$ w/s, $\theta=0.8$), on the 21.3% of questions where gold evidence is verbatim-present and BM25-retrievable, 95.2% are streamable under the operationally meaningful sufficiency criterion; on the remaining majority, a grounding-free top-1-settling fallback—which signals only that the lexical top-1 has settled, not that gold is in hand—yields 68.1%. The two combine to a 73.9% blend over the full benchmark, but they measure different things and the favorable suffi-

ciency figure is conditional on that 21.3% slice; because BM25 retrieves on early lexical anchors, that slice may itself bias stabilization early, but a dense-retriever replication (§5.5) shows the signal is not a lexical artifact—dense sufficiency stabilizes if anything later, not earlier. On the favorable slice, ϕ_{suf} is bracketed to $[0.26, 0.281]$ by exact and relaxed grounding—both early, though still conditional on the slice. Question type produces a statistically significant but small early/late split (Kruskal–Wallis $p = 0.017$, $\varepsilon^2=0.04$, $\sim 4\%$ of rank variance), a weak type-level tendency rather than a precise trigger rule.

1 Introduction

Tool-augmented dialogue systems increasingly rely on external retrieval to remain factual, but tool calls add latency that disrupts conversational flow. Streaming RAG (Arora et al., 2025) addresses this by predicting and issuing tool queries *in parallel with* the user’s still-arriving input, then reflecting on whether retrieved results suffice. The approach reports sizable accuracy and latency improvements and is modality-agnostic, applying equally to typed input.

The reported latency benefit, however, is an aggregate over a benchmark. It leaves a basic question unanswered: **for which queries can speculation help at all, and by how much?** The answer is not a property of the model, the retriever, or the speech stack—it is a property of *where, within an utterance, the information that determines the tool query first appears*. If the decisive term arrives last (“who makes the console called the *PlayStation*”), no early speculative query can retrieve the right evidence and the latency win collapses to zero regardless of system quality. If the decisive term arrives early, almost the entire tool latency can be hidden behind the remaining input. We initially expected these two regimes to show up as a *bimodal* distribution of stabilization; empirically (§5.1) the picture

is better described as a large early-stabilizing mass with a thin late tail—the early case is common and the late case is the rarer failure mode—and we report the distribution as found rather than as hypothesized.

We study this directly. By characterizing tool-intent stabilization as a measurable, query-intrinsic quantity, we obtain (i) a model-agnostic *aggregate* bound on the share of tool latency that can be hidden behind ongoing input, computable *before* any system is built; (ii) a principled account of the failure mode; (iii) a coarse build/skip signal—weak at the level of an individual query—for whether a learned trigger is worth its cost; and (iv) results that transfer across text and speech because the quantity is modality-agnostic. Our contribution is measurement and analysis rather than a new system.

2 Background and Related Work

Retrieval-Augmented Generation (Lewis et al., 2020) grounds generation in retrieved evidence; dense passage retrieval (Karpukhin et al., 2020) and the sparse BM25 model (Robertson and Zaragoza, 2009) are the standard retrieval backbones. Streaming RAG (Arora et al., 2025) adds a speculative dimension: a *Trigger* decides when to fire a tool query from a partial utterance, multiple speculative *Threads* run in parallel, and a *Reflector* judges whether results suffice. The idea is closely related in spirit to speculative decoding (Leviathan et al., 2023), which hides latency by doing work that may be discarded; here the discarded work is a premature tool call. Our analysis is orthogonal to the system: we quantify the headroom that any such system can exploit, set by the query alone.

Incremental comprehension and anticipation.

That the referential intent of an utterance is fixed incrementally—often well before the utterance ends—is among the most robust findings in psycholinguistics. Altmann and Kamide (1999) showed that a verb such as “eat” directs anticipatory attention to edible objects before they are named, so partial input already restricts what can follow. Kamide et al. (2003) extend this cross-linguistically: in head-final Japanese, case-marked pre-verbal arguments let comprehenders anticipate a forthcoming argument even before the verb (the head) is heard—*pre-head* anticipation (Kamide and Mitchell, 1999). This is the human analogue of firing a tool query from a prefix: *where* the disambiguating cue sits—varying with word order and

case morphology—governs how early intent stabilizes.

Stabilization over a growing prefix. The engineering counterpart—when a prediction over a growing input prefix becomes safe to commit—is the central question of *simultaneous* and *incremental* NLP. Simultaneous machine translation must decide, token by token, whether enough source has arrived to emit output: wait- k and prefix-to-prefix policies (Ma et al., 2019) and learned read/write agents (Gu et al., 2017) trade latency against the risk of committing before the disambiguating word arrives—precisely our *L*-versus-residual tension—and Grissom II et al. (2014) cast it as an agent that waits until it can confidently predict the sentence-final verb before committing, whose failure mode—premature commitment forcing revision—is exactly what our volatility V records. Closest to our measurement is incremental spoken language understanding: DeVault et al. (2011) build a meaning frame from partial ASR and quantify understanding as a function of prefix length, and the incremental-unit framework (Schlangen and Skantze, 2011) formalizes the same prefix-monotonicity and revision concerns. These efforts target a *closed* intent or dialogue-act ontology. Our contribution is to port the lens to *tool/retrieval* intent over an open-world knowledge benchmark: rather than learning a commit policy, we *measure* the query-intrinsic point at which the retrieval target stabilizes, which upper-bounds what any such policy could hide.

3 Problem Formalization

Let a query Q be a word sequence $q_{1:n}$ of length n words, and let $q_{1:t}$ denote the prefix of the first t words (we measure stabilization in words throughout, matching the whitespace tokenization used by both the retriever and the input-cadence model below). Let R be a retriever returning a ranked document list; write $d_t = \text{top-1}(R(q_{1:t}))$ for the top document retrieved from the prefix, d_n for the full-query top document, and d^* for the gold answer-bearing document.

Self-consistency stabilization t_{sc} . the smallest t such that for all $s \in \{t, \dots, n\}$, $d_s = d_n$: every prefix from t onward retrieves the same top document as the full query.

Sufficiency stabilization t_{suf} . the smallest t such that $d^* \in \text{top-}k(R(q_{1:t}))$: the prefix already surfaces the gold evidence. This is the operationally

meaningful notion, since the full query itself may retrieve imperfectly.

Stabilization fraction $\phi = t^*/n \in (0, 1]$, reported for both t_{sc} and t_{suf} . Lower ϕ means earlier stabilization and more headroom to hide latency.

Stabilization volatility V , the number of top-1 changes occurring *after* d_n is first reached. High V signals early-commit risk: a confident-looking early result that a later token would overturn.

Hidden latency.

$$H(Q; L, \delta) = \min\left(L, \max\left(0, \frac{n-t^*}{\delta}\right)\right), \quad (1)$$

where L is tool latency (ms) and δ is the input cadence in words per second, so that $(n - t^*)/\delta$ is the residual input time (in seconds, scaled to ms) still to arrive after the query has stabilized. H is the portion of tool latency that completes behind the user’s remaining input. It bounds the *input-hideable* component of the per-query perceived-latency saving—not the total saving, since a real pipeline can additionally overlap other costs (§5.4). A slower cadence (smaller δ) lengthens the residual and so raises H when tool latency is the larger term.

Streamable. $\text{Streamable}(Q; L, \delta, \theta)$ holds iff $H \geq \theta L$ for a coverage threshold θ (e.g. $\theta = 0.8$ hides at least 80% of tool latency). The streamable fraction of a workload is the central deployment-relevant statistic.

Research questions. We address four pre-registered questions. **RQ1** asks how ϕ is distributed across queries; we expect a bimodal shape with an early-stabilizing mass and a late-stabilizing tail. **RQ2** asks what fraction of queries admit latency hiding as L , δ , and θ vary; we expect that for large L the binding constraint is the residual input time $(n - t^*)/\delta$, so a *slower* cadence paradoxically *raises* the streamable fraction. **RQ3** asks whether the H bound matches a working pipeline. **RQ4** asks which query features predict early vs. late stabilization; we expect question type to be the dominant signal, with simple lookups stabilizing earlier than multi-hop or aggregation queries. We address RQ1–RQ4 in §5 and include top- k and dense-retriever robustness arms (Table 1, §5.5) as sensitivity checks. A cross-linguistic condition is left to future work. We report disconfirmations alongside confirmations: RQ2 is confirmed, but the RQ1 distribution is right-skewed rather than

bimodal (§5.1), and the RQ4 ordering is governed by entity position rather than reasoning complexity (§5.2).

4 Method

Data. We use CRAG, the Comprehensive RAG Benchmark (Yang et al., 2024), Task 1&2 dev set: 2706 questions, of which 1371 are the validation split we analyze. Each question carries up to five retrieved web pages, a gold answer (plus alternates), and a question-type label.

Grounding d^* . CRAG provides no gold-*passage* label, only gold answer strings. We derive d^* by grounding the normalized gold answer(s) into passage text: substring match for multi-token answers, and word-boundary match for single-token answers, over the answer and its alternates. Questions whose answer is never a literal span (false-premise, “I don’t know”, and many aggregation/dynamic answers) are *ungroundable* and excluded from t_{suf} ; we report the groundable rate explicitly. Because t_{sc} requires no grounding, we report it as a grounding-free robustness check alongside t_{suf} .

String grounding is a recall-oriented heuristic and can produce false positives: a single-token gold answer (a year, a number, “yes”/“no”) may match a passage *coincidentally*, registering sufficiency before the genuinely answer-bearing evidence arrives and biasing t_{suf} —and thus ϕ_{suf} —*early*. Multi-token answers, which require a contiguous substring match, are far less prone to this. We therefore read ϕ_{suf} as a lower bound on stabilization time on the groundable subset, and treat the early-stabilization finding as directionally robust rather than exact.

To bound the population-selection bias this strictness induces, we also run a *relaxed* grounding arm (§5.5): for multi-token answers it accepts a passage if all of the answer’s content tokens (stopwords dropped) appear anywhere in it—bag-of-content-token containment, dropping only the contiguity requirement. This is deliberately higher-recall and lower-precision than the substring rule, so the two arms *bracket* the true groundable population and the ϕ_{suf} that goes with it.

Passages and retrieval. Page HTML is cleaned to text and chunked into 120-word windows with a 20-word overlap (stride 100), capped at 400 chunks per question. We index each question’s passages with a pure-Python lexical BM25 (Robertson and

Zaragoza, 2009) ($k_1=1.5$, $b=0.75$) and retrieve over every prefix $q_{1:t}$, $t = 1 \dots n$, recording d_t and the top- k set at each prefix. We sweep $k \in \{1, 3, 5\}$ for the sufficiency definition. All retrieval is deterministic; the analysis uses the CRAG dev_v4 snapshot (the released web pages, so retrieval is fixed and not subject to live drift). The two stochastic or selection steps are seeded and reproducible: the 60-question RQ3 subset is simply the first 60 retrieved-gold questions in dataset order (no sampling), and the bootstrap confidence intervals (§5.2) use 10,000 resamples at seed 0.

Latency model and harness (RQ3). We stream each query as a uniform word sequence—a controlled proxy for input cadence δ —and sweep (L, δ, θ) analytically over the grid $L \in \{100, 300, 600, 1000\}$ ms, $\delta \in \{2, 3, 4\}$ w/s, $\theta \in \{0.5, 0.8, 1.0\}$ to obtain the streamable surface (RQ2). For RQ3 we replay a subset of questions through a faithful asynchronous Streaming RAG harness (fixed-interval Trigger, parallel Threads, Reflector) calibrated with the per-stage latencies reported by Arora et al. (2025) (Table 3): query generation ≈ 590 ms and fuse/reflect ≈ 2520 ms, in addition to the swept tool latency L . We compare the *measured* perceived-latency saving (baseline minus streaming) against the H bound.

Reproducibility. The pipeline is training-free and CPU-only. Code and the derived d^* passage labels (the only non-trivial artifact, since CRAG ships none) are available at [anonymized]; every per-question metric and the full (L, δ, θ) grid can be recomputed offline without re-running retrieval.

5 Results

We analyze the 1371 validation questions of CRAG Task 1&2. Under our string-grounding procedure, 35.4% of questions are *groundable* (the gold answer appears verbatim in the retrieved pages); the remainder—dominated by false-premise items and answers that never occur as a literal span (many aggregation and dynamic questions)—are excluded from sufficiency statistics. At $k=3$, BM25 surfaces the gold passage in the top- k at some prefix for 21.3% of all questions. All reported ϕ_{suf} statistics are conditional on this retrieved-gold population; ϕ_{sc} is defined for every question and serves as a grounding-free check.

This retrieved-gold subset is not a random sam-

ple of the benchmark: it is the intersection of questions whose answer is verbatim-groundable *and* whose gold passage BM25 can surface from some prefix. Both filters plausibly select for queries with a salient, early lexical anchor, so the conditional ϕ_{suf} statistics should be read as characterizing this favorable slice, not “most answerable questions”; if anything, the selection biases the reported stabilization *earlier* than it would be on the full workload. We therefore lean on ϕ_{sc} , defined on all 1371 questions, as the grounding-free companion to every sufficiency claim, and we bound the magnitude of the selection effect with a relaxed-grounding arm below (§5.5): recovering a sixth more of the population moves ϕ_{suf} by under 0.02, so the early-stabilization finding survives the expansion rather than dissolving into it.

5.1 RQ1: How is stabilization distributed?

The two stabilization notions behave very differently (Fig. 1). Self-consistency stabilizes *late*: the lexical top-1 keeps changing until, on average, ϕ_{sc} reaches mean 0.67 (median 0.73)—roughly three-quarters of the way through the query. Sufficiency, in contrast, stabilizes *early*: the gold passage enters the top- k at mean $\phi_{\text{suf}} = 0.26$ and median 0.14, bracketed to $[0.26, 0.281]$ by a relaxed-grounding arm (§5.5)—both early. The gap is the central empirical observation of this study: **the answer-bearing document is retrievable from a short prefix long before the query’s lexical top-1 settles**. For streaming tool use this is the favorable regime—one need not wait for the utterance to “finish” for the right evidence to be in hand.

The ϕ_{suf} distribution (Fig. 1) is right-skewed with a large mass near zero and a thinner late tail, rather than cleanly bimodal as H1 anticipated. Volatility is modest: the post-stabilization top-1 changes at all for only 20.9% of questions (mean $V = 0.52$), so early-commit risk is concentrated in a minority of queries rather than pervasive.

5.2 RQ4: What predicts early vs. late stabilization?

Question type produces a coarse early/late split in ϕ_{suf} , spanning a roughly threefold range across medians (Fig. 2). A Kruskal–Wallis test across all eight types rejects equality of the ϕ_{suf} distributions ($H=17.0$, $df=7$, $p=0.017$); restricting to the five classes with $n \geq 10$ —which carry the effect—the test is stronger ($H=13.9$, $df=4$, $p=0.008$). The effect, though significant, is *small*: the rank-based ef-

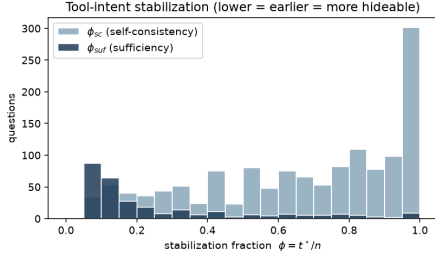


Figure 1: Distribution of stabilization fraction ϕ at $k=3$. Sufficiency (ϕ_{suf} , dark) concentrates near zero—gold evidence is retrievable early—while self-consistency (ϕ_{sc} , light) is late and broad.

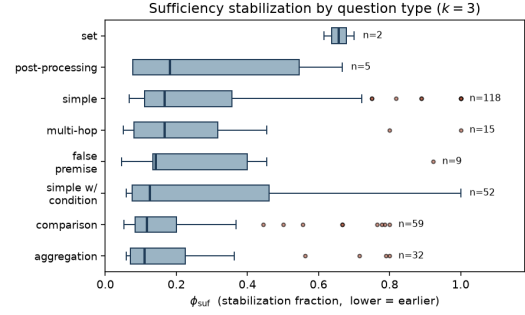


Figure 2: Distribution of ϕ_{suf} by question type ($k=3$, sorted by median). The $3\times$ range across medians is real, but box overlaps are large for adjacent types—only the extremes (*aggregation/comparison* earliest, *set* latest) are reliably ordered. Ordering tracks entity position, not reasoning complexity.

fect size is $\varepsilon^2=0.04$ (both for all eight types and for the five large classes), i.e. question type explains on the order of 4% of the rank variance in ϕ_{suf} . Consistent with a small effect, the interquartile ranges in Fig. 2 overlap substantially for adjacent types: the ordering is reliable only at the extremes (*aggregation* and *comparison* earliest, *set* latest—*set* confirmed at $n=20$ under the relaxed-grounding arm, §5.5), not as a strict cell-by-cell ranking, and *post-processing* ($n=5$) has too few points to place reliably. A Dunn’s post-hoc with Holm correction makes this precise: *no* pairwise contrast survives at $\alpha=0.05$ (the closest, *simple* vs. *aggregation*, reaches $p_{\text{Holm}}=0.05$). The omnibus effect is therefore carried by the overall rank distribution, not by any single separable pair. We accordingly claim a significant but coarse type-level effect—a robust early/late *tendency*, not a precise per-type order. The qualitative ordering refines hypothesis H3 rather than confirming it. H3 expected simple lookups to stabilize early and aggregation/comparison/multi-hop to stabilize late. We instead find *aggregation* and *comparison* among the *earliest*-stabilizing types, with *set* questions latest. The explanation is that sufficiency measures when the gold *document* becomes retrievable, not when the answer can be *computed*: a comparison or aggregation question often names its key entities up front, so the answer-bearing page is retrievable early even though producing the final answer requires post-retrieval reasoning. In other words, retrieval-sufficiency stabilization is governed by entity position in the utterance, which is not aligned with reasoning complexity. This entity-position account is carried by the well-populated classes (*simple*, *comparison*, *aggregation*, *simple_w_condition*), so it does not hinge on the noisy small- n cells. We

present it as an explanation motivated by the data rather than one we have isolated: it yields a directly falsifiable prediction—that the word position of the first named entity should track t_{suf} more tightly than question type does—which a lightweight entity-tagging pass over the queries could test without any new retrieval. If that correlation held, first-entity position would also furnish a far cheaper deployment signal than the full prefix-retrieval sweep. We flag both as the natural next measurement.

5.3 RQ2: What fraction of queries admit latency hiding?

Using $t^* = t_{\text{suf}}$ where available and falling back to t_{sc} otherwise, we compute the streamable fraction over the full (L, δ, θ) grid (Fig. 3). At the central operating point ($L=600$ ms, $\delta=3$ w/s, $\theta=0.8$), 73.9% of queries admit hiding at least 80% of tool latency.

This blended figure mixes two populations and two notions of stabilization, and we report it decomposed. On the 21.3% of queries with retrieved gold—where $t^* = t_{\text{suf}}$ reflects evidence actually in hand—95.2% are streamable; on the remaining 78.7%, where we fall back to the grounding-free t_{sc} (top-1 stopped moving), 68.1% are. The sufficiency-only number, 95.2%, is the one that matches our framing and is the honest headline for “evidence is retrievable in time”; the 73.9% blend is lower because it is dominated by the t_{sc} majority, for which “streamable” means only that the lexical top-1 has settled, not that gold is in the top- k . Readers who care about evidence sufficiency should weight the 95.2% figure; those who

treat top-1 stability as the deployment signal get the blend.

The blend also counts only upside. “Streamable” asks whether enough tool latency *can* be hidden; it is silent on whether the speculative query was *right*, and H ’s $\max(0, \cdot)$ floor cannot represent the negative saving a mis-fire incurs (§5.4). One might expect this blind spot to bite hardest in the t_{sc} -fallback majority, which stabilizes *late* (mean $\phi=0.68$ vs. ≈ 0.29 on the retrieved-gold replay sample), giving a fixed-interval trigger more room to fire before intent settles. We tested this by replaying 60 fallback questions (the harness previously replayed only retrieved-gold) through the same pipeline at $L=600$ ms. The expectation does *not* hold: the net-negative-saving rate is 1.7% (1/60), identical to the 1.7% on the favorable retrieved-gold population, and mean savings are likewise comparable. The reason is the same constant that dominates RQ3 (§5.4): the query-generation overlap is hidden regardless of whether the speculative *retrieval* was right, so late stabilization lowers H without proportionally raising the mis-fire rate at this operating point. Mis-fires are thus real but rare ($\approx 1.7\%$) across both populations here; the streamable fraction should still be read as an upper envelope on benefit paired with this small measured downside, and the rate may grow at larger L or with a more aggressive trigger—a sweep we leave to future work.

Two further patterns stand out. First, for small tool latency ($L \leq 300$ ms) the constraint is essentially never binding: 82.6% of queries are streamable regardless of cadence—the residual input time trivially covers a short tool call. Second, and confirming **H2**, when tool latency is large ($L=1000$ ms) the binding constraint becomes residual input time, so a *slower* cadence *raises* the streamable fraction: from 54.5% at $\delta=4$ w/s up to 73.9% at $\delta=2$ w/s. Faster typists and faster talkers leave less room to hide a slow tool.

5.4 RQ3: Does the bound match a working pipeline?

We replayed the same 60 retrieved-gold questions through the asynchronous Streaming RAG harness at $\delta=3$ w/s and *two* tool latencies, $L \in \{600, 1000\}$ ms (Fig. 4). In the aggregate the bound is *conservative* at both: measured perceived-latency savings (baseline minus streaming) average 1122 ms vs. the 590 ms H bound at $L=600$, and 1457 ms vs. 950 ms at $L=1000$. The overshoot is expected and informative: H models only the tool

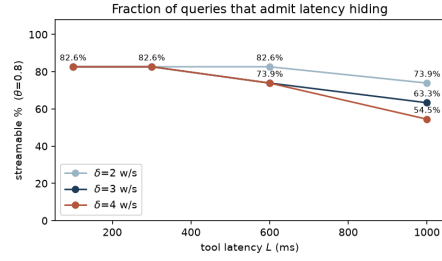


Figure 3: Streamable fraction vs. tool latency L at $\theta=0.8$, one line per cadence δ , with exact values annotated. The curves separate only once L is large enough for residual input time to bind; there, slower cadence dominates (H2).

latency hidden behind input, whereas the calibrated pipeline additionally overlaps the LLM’s query-generation step (≈ 590 ms (Arora et al., 2025)) with the user’s input. Indeed the overshoot is almost exactly that step: at $L=600$ the bound mean (590 ms) plus the query-generation constant (≈ 590 ms) recovers the measured mean (≈ 1122 ms) to within noise—a useful sanity check, but also a caution that this paragraph mostly re-derives H plus a constant rather than independently testing H (we address that next). So a query that H deems streamable will, in practice, save *at least* as much as predicted—the property that makes H safe to plan with.

We are deliberately careful about the stronger claim the bound does *not* support: per-query prediction. At $L=600$ the bound saturates—59 of the 60 queries share $H=600$ ms (the input residual exceeds L)—leaving no horizontal spread to test ranking, and the rank correlation between H and the measured saving is negligible (Spearman $\rho=+0.14$). We therefore re-ran the *identical* questions at $L=1000$ specifically to break the saturation: H now takes three distinct values (0, 667, 1000 ms) with 7 of 60 queries below the cap. The spread appears, but the correlation does not: $\rho=+0.12$ over all 60, and exactly 0.00 within the unsaturated tail. Two structural reasons explain why raising L cannot rescue per-query prediction. First, the dominant component of realized saving—the constant ≈ 590 ms query-generation overlap—is independent of H , so it adds a near-uniform offset that swamps the cross-query signal H encodes. Second, the failure mode is *negative*, and H ’s $\max(0, \cdot)$ floor cannot represent it. RQ3 thus confirms H as a conservative aggregate bound but *not* a per-query ranker; a predictive model would need to add the

query-generation overlap to the hideable budget and replace the floor with a mis-fire penalty.¹

The negative tail makes the second point concrete and reconnects volatility V to outcomes. The single query that *loses* time—a 17-word comparison ($t^*=13$)—costs -543 ms at $L=600$ and -940 ms at $L=1000$, yet its H is *maximal* (1000 ms): with $n - t^*=4$ words of nominal residual the bound rates it fully streamable while the fixed-interval trigger, firing before intent settles at word 13, mis-retrieves and pays a re-fire penalty. Notably this query has $V=0$, so volatility did *not* flag it—the risk here is *late* stabilization under an early-firing trigger, not post-stabilization churn. This is a single-query observation ($n=1$ net-negative across the replayed populations), so we offer the reading as a mechanism-level intuition rather than a validated design rule: it *suggests* that an early-commit safeguard should gate on ϕ (how late intent settles relative to the trigger cadence) at least as much as on V . Whether ϕ in fact out-predicts V for mis-fires requires a larger negative sample—a trigger-aggressiveness and large- L sweep that deliberately induces mis-fires—which we leave to future work and do not claim to have established here.

5.5 Robustness

Grounding (bounding the selection bias). Our sharpest claims live on the 21.3% retrieved-gold slice, whose selection plausibly biases stabilization early (§5.1). To bound that bias we re-derive d^* with a deliberately *relaxed* matcher—bag-of-content-token containment rather than contiguous substring (§4)—which trades precision for recall and so expands the groundable population. Groundability rises from 35.4% to 43.3% and the retrieved-gold rate from 21.3% to 24.4% (a sixth more questions). If the early-stabilization result were largely a selection artifact, admitting these harder-to-ground questions should pull ϕ_{suf} up sharply. It does not: mean ϕ_{suf} moves only from 0.26 to 0.281 (median 0.143 $\rightarrow$ 0.154). The true value is thus bracketed in $[0.26, 0.281]$ —both early—and

¹Formalizing the first correction without re-running anything: a tighter aggregate bound $H' = \min(L, \text{residual}) + \min(t_{\text{ag}}, \text{residual})$ folds the query-generation step ($t_{\text{qg}} \approx 590$ ms), which the pipeline overlaps with input regardless of retrieval correctness, into the hideable budget. At $L=600$ this predicts $\approx 590 + 590 \approx 1180$ ms against the measured 1122 ms—recovering the overshoot to within noise. H' is still aggregate, not per-query: it adds a near-constant offset and so cannot rescue the rank correlation, and it inherits the $\max(0, \cdot)$ floor that the negative tail violates.

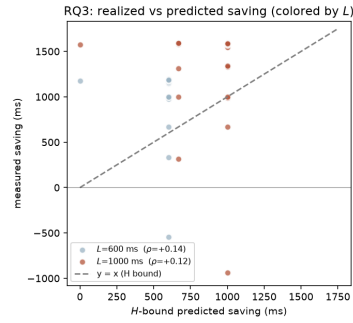


Figure 4: RQ3: per-question measured saving vs. the H bound for the same 60 questions, colored by tool latency L . Most points lie above $y=x$ because the pipeline also hides query-generation latency, which H does not model. At $L=600$ ms (light) the bound saturates—59/60 points share $H=600$ ms; at $L=1000$ ms (dark) H spreads across $\{0, 667, 1000\}$ ms, yet measured savings still do not track it (Spearman $\rho = +0.12$). The lone point below zero is a late-stabilizing query whose mis-fire the floored bound cannot express—and whose H is nonetheless maximal.

Table 1: Top- k robustness. ϕ_{sc} (mean 0.67 / median 0.73) is independent of k and omitted.

k	retrieved-gold	mean ϕ_{suf}	median ϕ_{suf}
1	15.6%	0.327	0.200
3	21.3%	0.264	0.143
5	25.0%	0.248	0.125

the by-type ordering is preserved, now on firmer footing: `set`, the latest type, grows from $n=2$ to $n=20$ and remains latest. We read the early-sufficiency finding as signal that survives a $1.5\times$ population expansion, not as an artifact of the strict matcher. (Fuzzy grounding is lower-precision, so it if anything biases ϕ_{suf} *down*; that the estimate barely moves is the reassuring direction.)

Retriever (top- k). Varying the sufficiency top- k shifts absolute values but preserves the qualitative picture (Table 1). Larger k surfaces the gold passage for more questions (retrieved-gold rate 15.6% $\rightarrow$ 25.0%) and stabilizes slightly earlier (mean ϕ_{suf} 0.327 $\rightarrow$ 0.248), as expected since a more permissive top- k is easier to satisfy. The self-consistency curve is unchanged by k (it depends only on the top-1), and the central streamable fraction is stable near 73–74%. The early-sufficiency / late-self-consistency gap holds at every k .

Retriever (dense). Our sharpest concern is that BM25’s reliance on early lexical anchors—

not a property of the queries—drives the low ϕ_{suf} we report. We test this directly by re-running the full sweep with a dense bi-encoder (all-MiniLM-L6-v2, cosine over the same passages) in place of BM25. The early-stabilization finding is not a lexical artifact: under dense retrieval sufficiency stabilizes if anything *later*, not earlier (mean ϕ_{suf} 0.26 $\rightarrow$ 0.31, median 0.14 $\rightarrow$ 0.22; ϕ_{sc} 0.67 $\rightarrow$ 0.85), while the dense retriever surfaces gold for *more* questions (retrieved-gold 21.3% $\rightarrow$ 26.5%). A semantic retriever that does not key on early keywords thus still finds the answer-bearing passage well before the query ends—median ϕ_{suf} = 0.22, within the first quarter of the utterance. The by-type ordering is preserved (Spearman ρ =0.79 between the BM25 and dense ϕ_{suf} rankings): *aggregation* and *comparison* remain among the earliest under both, and the disagreement is confined to the small- n / grounding-ambiguous cells (*false_premise*, *set*, *post-processing*). We read the dense arm as converting our principal open confound into a positive robustness result: the early-sufficiency signal is retriever-general, and BM25, if anything, is the *more* optimistic backbone.

6 Conclusion

Tool-intent stabilization is a query-intrinsic, model-agnostic quantity that bounds when streaming tool use can help and by how much. On CRAG, sufficiency stabilizes early on the groundable, retrievable subset (mean ϕ_{suf} bracketed to [0.26, 0.281] by exact and relaxed grounding, both early); question type produces a significant but coarse early/late split (Kruskal–Wallis p = 0.017, ε^2 =0.04); and the H bound is a conservative aggregate bound with realized savings meeting or exceeding it, at a streamable fraction of 73.9% across the full benchmark. Two limitations temper these claims, and we probed both rather than merely noting them. First, the sufficiency results are conditional on a favorable 21.3% slice; the relaxed-grounding arm brackets the population effect to under 0.02 in ϕ_{suf} (§5.5), bounding the finding as signal rather than artifact—though a fuller recovery (dense or learned grounding) remains open. Second, our system validation establishes conservativeness in aggregate but *not* per-query prediction: at L =1000 ms, realized savings do not track H (Spearman ρ =+0.12), because a constant query-generation overlap dominates the saving and trigger mis-fires produce neg-

ative outcomes the floored bound cannot represent (§5.4). The downside is real but rare ($\approx$ 1.7% net-negative at L =600). Closing the prediction gap—adding the query-generation overlap and a mis-fire penalty—together with a regression of ϕ on query features are the natural next steps. Even so, the present results give system builders a principled, training-free basis for trigger design and a build/skip decision for a learned trigger.

Limitations

Input timing. A uniform word stream is a proxy for speech; real ASR pacing is variable and partial hypotheses get revised. We bound this here by using clean text and treat speech timing as planned follow-up.

Stabilization measure. Top-1 equality is a coarse proxy for “intent settled.” We mitigate by also reporting sufficiency against the gold document and by varying top- k .

Grounding. d^* is derived by string matching, so the groundable population excludes non-verbatim answers; reported ϕ_{suf} statistics are conditional on groundability, and we report t_{sc} as a grounding-free check.

Fallback population. On the 78.7% of queries where BM25 never surfaces the gold passage from any prefix, stabilization is measured only by t_{sc} (when the top-1 stops moving). If the top-1 in this population is the wrong document, early settling is *stable mis-retrieval*, not evidence that the speculative query was right: t_{sc} certifies that intent has *stopped changing*, not that it is *correct*. This is the central gap between our measurement and end-to-end streaming-RAG quality, and it is why we keep the sufficiency and self-consistency populations separate throughout (§5.3) rather than reading the blended streamable fraction as an evidence-quality claim. Our RQ3 replay finds the realized-saving and net-negative rates comparable across the two populations at L =600 (§5.3), but that probes *perceived latency*, not retrieval correctness, on the fallback set.

Retriever and benchmark. BM25 is lexical and phrasing-sensitive, and is known to retrieve on early topical anchors. Because t_{suf} asks only when the gold page enters the top- k , that early-anchor behavior could itself drive the low ϕ_{suf} we report. We addressed this head-on with the dense-retriever

arm (§5.5): a bi-encoder that weights terms semantically rather than lexically surfaces gold for *more* questions yet stabilizes if anything *later*, and the by-type ordering is preserved ($\rho=0.79$)—so the early-stabilization claim is retriever-general, not a BM25 artifact. What remains open is benchmark generality: a single English QA set leaves cross-dataset and cross-lingual replication—where verb-final word order may shift t_{suf} under lexical but not dense retrieval—as the natural next step, not optional polish.

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